GNU nano 8.7.1

```c
#include <stdio.h>
#define ZONES 5
#define DAYS 7
const char *zoneNames[ZONES] = {
    "Academic Block",
    "Hostel",
    "Library",
    "Cafeteria",
    "Sports Area"
};
const char *dayNames[DAYS] = {
    "Day 1", "Day 2", "Day 3", "Day 4",
    "Day 5", "Day 6", "Day 7"
};
/* Required function prototypes */
void inputData(float temp[ZONES][DAYS],
               float humidity[ZONES][DAYS],
               int aqi[ZONES][DAYS]);
void displayData(float temp[ZONES][DAYS],
                 float humidity[ZONES][DAYS],
                 int aqi[ZONES][DAYS]);
void calculateAverages(float temp[ZONES][DAYS],
                       float humidity[ZONES][DAYS],
                       int aqi[ZONES][DAYS],
                       float avgTemp[ZONES],
                       float avgHumidity[ZONES],
                       float avgAQI[ZONES]);
void detectRiskDays(float temp[ZONES][DAYS],
                    float humidity[ZONES][DAYS],
                    int aqi[ZONES][DAYS],
                    int riskCount[ZONES],
                    int riskDays[ZONES][DAYS]);
void computeGreenIndex(float avgTemp[ZONES],
                       float avgHumidity[ZONES],
                       float avgAQI[ZONES],
                       float greenIndex[ZONES]);
void displayRanking(float greenIndex[ZONES]);
void displayRecommendations(float avgTemp[ZONES],
                            float avgHumidity[ZONES],
                            float avgAQI[ZONES],
                            int riskCount[ZONES]);
/* Input and validate all readings */
void inputData(float temp[ZONES][DAYS],
               float humidity[ZONES][DAYS],
               int aqi[ZONES][DAYS])
{
    int z, d;
    for (z = 0; z < ZONES; z++) {
        printf("\n--- Enter readings for %s ---\n", zoneNames[z]);
        for (d = 0; d < DAYS; d++) {
            printf("\n%s\n", dayNames[d]);
            do {
                printf("Temperature (10-50 C): ");
                scanf("%f", &temp[z][d]);
```

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^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

26°C

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```
            if (temp[z][d] < 10 || temp[z][d] > 50)
                printf("Invalid temperature. Try again.\n");
        } while (temp[z][d] < 10 || temp[z][d] > 50);
        do {
            printf("Humidity (0–100%%): ");
            scanf("%f", &humidity[z][d]);
            if (humidity[z][d] < 0 ||
                humidity[z][d] > 100)
                printf("Invalid humidity. Try again.\n");
        } while (humidity[z][d] < 0 ||
                 humidity[z][d] > 100);
        do {
            printf("AQI (0–500): ");
            scanf("%d", &aqi[z][d]);
            if (aqi[z][d] < 0 || aqi[z][d] > 500)
                printf("Invalid AQI. Try again.\n");
        } while (aqi[z][d] < 0 || aqi[z][d] > 500);
    }
}
printf("\nAll readings saved successfully.\n");
}
/* Display the three data matrices */
void displayData(float temp[ZONES][DAYS],
                 float humidity[ZONES][DAYS],
                 int aqi[ZONES][DAYS])
{
    int z, d;
    printf("\nTEMPERATURE MATRIX (C)\n");
    printf("%-18s", "Zone");
    for (d = 0; d < DAYS; d++)
        printf("%10s", dayNames[d]);
    printf("\n");
    for (z = 0; z < ZONES; z++) {
        printf("%-18s", zoneNames[z]);
        for (d = 0; d < DAYS; d++)
            printf("%10.1f", temp[z][d]);
        printf("\n");
    }
    printf("\nHUMIDITY MATRIX (%%)\n");
    printf("%-18s", "Zone");
    for (d = 0; d < DAYS; d++)
        printf("%10s", dayNames[d]);
    printf("\n");
    for (z = 0; z < ZONES; z++) {
        printf("%-18s", zoneNames[z]);
        for (d = 0; d < DAYS; d++)
            printf("%10.1f", humidity[z][d]);
        printf("\n");
    }
    printf("\nAQI MATRIX\n");
    printf("%-18s", "Zone");
    for (d = 0; d < DAYS; d++)
        printf("%10s", dayNames[d]);
    printf("\n");
```

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```
    for (z = 0; z < ZONES; z++) {
        printf("%-18s", zoneNames[z]);
        for (d = 0; d < DAYS; d++)
            printf("%10d", aqi[z][d]);
        printf("\n");
    }
}
/* Calculate average temperature, humidity, and AQI */
void calculateAverages(float temp[ZONES][DAYS],
                       float humidity[ZONES][DAYS],
                       int aqi[ZONES][DAYS],
                       float avgTemp[ZONES],
                       float avgHumidity[ZONES],
                       float avgAQI[ZONES])
{
    int z, d;
    float sumTemp, sumHumidity, sumAQI;
    printf("\nZONE-WISE AVERAGES\n");
    printf("%-18s %12s %12s %12s\n",
           "Zone", "Temp (C)", "Humidity", "AQI");
    for (z = 0; z < ZONES; z++) {
        sumTemp = 0;
        sumHumidity = 0;
        sumAQI = 0;
        for (d = 0; d < DAYS; d++) {
            sumTemp += temp[z][d];
            sumHumidity += humidity[z][d];
            sumAQI += aqi[z][d];
        }
        avgTemp[z] = sumTemp / DAYS;
        avgHumidity[z] = sumHumidity / DAYS;
        avgAQI[z] = sumAQI / DAYS;
        printf("%-18s %12.2f %12.2f %12.2f\n",
               zoneNames[z],
               avgTemp[z],
               avgHumidity[z],
               avgAQI[z]);
    }
}
/* Count each risk day only once */
void detectRiskDays(float temp[ZONES][DAYS],
                    float humidity[ZONES][DAYS],
                    int aqi[ZONES][DAYS],
                    int riskCount[ZONES],
                    int riskDays[ZONES][DAYS])
{
    int z, d;
    printf("\nRISK DAY REPORT\n");
    for (z = 0; z < ZONES; z++) {
        riskCount[z] = 0;
        printf("\n%s:\n", zoneNames[z]);
        for (d = 0; d < DAYS; d++) {
            riskDays[z][d] = 0;
            if (aqi[z][d] > 150 ||
```

```
            temp[z][d] > 35 ||
            humidity[z][d] > 80) {
            riskDays[z][d] = 1;
            riskCount[z]++;
        }
    }
    printf("Risk days: %d\n", riskCount[z]);
    if (riskCount[z] == 0) {
        printf("No risk days detected.\n");
    } else {
        printf("Risk detected on: ");
        for (d = 0; d < DAYS; d++) {
            if (riskDays[z][d] == 1)
                printf("%s ", dayNames[d]);
        }
        printf("\n");
    }
  }
}
/* Calculate Green Index using the assignment formula */
void computeGreenIndex(float avgTemp[ZONES],
                       float avgHumidity[ZONES],
                       float avgAQI[ZONES],
                       float greenIndex[ZONES])
{
    int z;
    float aqiScore, tempScore, humidityScore;
    printf("\nGREEN INDEX REPORT\n");
    for (z = 0; z < ZONES; z++) {
        aqiScore = 100 - (avgAQI[z] / 5);
        tempScore = 100 - (4 * (avgTemp[z] - 25));
        humidityScore = 100 - (2 * (avgHumidity[z] - 50));
        /* Clamp negative scores to zero */
        if (aqiScore < 0)
            aqiScore = 0;
        if (tempScore < 0)
            tempScore = 0;
        if (humidityScore < 0)
            humidityScore = 0;
        greenIndex[z] =
            (0.4 * aqiScore) +
            (0.3 * tempScore) +
            (0.3 * humidityScore);
        printf("%-18s Green Index: %.2f\n",
               zoneNames[z], greenIndex[z]);
    }
}
/* Display zones in descending Green Index order */
void displayRanking(float greenIndex[ZONES])
{
    int i, j, index;
    int used[ZONES] = {0};
    printf("\nZONE RANKING (BEST TO WORST)\n");
    printf("%-8s %-18s %s\n", "Rank", "Zone", "Green Index");
```

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```
    for (i = 0; i < ZONES; i++) {
        index = -1;
        for (j = 0; j < ZONES; j++) {
            if (!used[j] &&
                (index == -1 ||
                 greenIndex[j] > greenIndex[index])) {
                index = j;
            }
        }
        used[index] = 1;
        printf("%-8d %-18s %.2f\n",
               i + 1, zoneNames[index], greenIndex[index]);
    }
}
/* Generate recommendations based on zone averages and risk days */
void displayRecommendations(float avgTemp[ZONES],
                            float avgHumidity[ZONES],
                            float avgAQI[ZONES],
                            int riskCount[ZONES])
{
    int z;
    printf("\nENVIRONMENTAL RECOMMENDATIONS\n");
    for (z = 0; z < ZONES; z++) {
        printf("\n%s:\n", zoneNames[z]);
        if (avgAQI[z] > 150)
            printf("- Reduce vehicle idling and improve ventilation.\n");
        if (avgTemp[z] > 35)
            printf("- Plant shade trees or install green cover.\n");
        if (avgHumidity[z] > 80)
            printf("- Improve ventilation and check for stagnant water.\n");
        if (riskCount[z] > 0)
            printf("- Monitor environmental conditions regularly.\n");
        if (avgAQI[z] <= 150 &&
            avgTemp[z] <= 35 &&
            avgHumidity[z] <= 80 &&
            riskCount[z] == 0) {
            printf("- Maintain current environmental conditions.\n");
        }
    }
}
int main(void)
{
    float temp[ZONES][DAYS];
    float humidity[ZONES][DAYS];
    int aqi[ZONES][DAYS];
    float avgTemp[ZONES];
    float avgHumidity[ZONES];
    float avgAQI[ZONES];
    float greenIndex[ZONES];
    int riskCount[ZONES];
    int riskDays[ZONES][DAYS];
    int choice;
    int dataLoaded = 0;
    do {
```

^G Help ^X Exit ^O Write Out ^R Read File ^F Where Is ^\ Replace ^K Cut ^U Paste ^T Execute ^J Justify ^C Locat ^/ Go To

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GNU nano 8.7.1

```
        printf("1. Input / Update Data\n");
        printf("2. Display Data\n");
        printf("3. Zone-wise Averages\n");
        printf("4. Detect Risk Days\n");
        printf("5. Green Index and Ranking\n");
        printf("6. Recommendations\n");
        printf("7. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        if (choice >= 2 && choice <= 6 && !dataLoaded) {
            printf("\nPlease enter the data first using Option 1.\n");
            continue;
        }
        switch (choice) {
            case 1:
                inputData(temp, humidity, aqi);
                dataLoaded = 1;
                break;
            case 2:
                displayData(temp, humidity, aqi);
                break;
            case 3:
                calculateAverages(temp, humidity, aqi,
                                  avgTemp, avgHumidity, avgAQI);
                break;
            case 4:
                detectRiskDays(temp, humidity, aqi,
                               riskCount, riskDays);
                break;
            case 5:
                calculateAverages(temp, humidity, aqi,
                                  avgTemp, avgHumidity, avgAQI);
                computeGreenIndex(avgTemp, avgHumidity,
                                  avgAQI, greenIndex);
                displayRanking(greenIndex);
                break;
            case 6:
                calculateAverages(temp, humidity, aqi,
                                  avgTemp, avgHumidity, avgAQI);
                detectRiskDays(temp, humidity, aqi,
                               riskCount, riskDays);
                displayRecommendations(avgTemp, avgHumidity,
                                       avgAQI, riskCount);
                break;
            case 7:
                printf("\nExiting EcoGuard. Goodbye!\n");
                break;
            default:
                printf("\nInvalid choice. Please try again.\n");
        }
    } while (choice != 7);
    return 0;
}
```

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^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To

yokshita@yokshithachowdary

```
yokshita@yokshithachowdary:~$ pwd
/home/yokshita
yokshita@yokshithachowdary:~$ cd
yokshita@yokshithachowdary:~$ nano p10.c
yokshita@yokshithachowdary:~$ gcc p10.c
yokshita@yokshithachowdary:~$ ./a.out

========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
Enter your choice: 1

--- Enter readings for Academic Block ---

Day 1
Temperature (10-50 C): 12
Humidity (0-100%): 23
AQI (0-500): 34

Day 2
Temperature (10-50 C): 23
Humidity (0-100%): 34
AQI (0-500): 456

Day 3
Temperature (10-50 C): 16
Humidity (0-100%): 34
AQI (0-500): 432

Day 4
Temperature (10-50 C): 45
Humidity (0-100%): 32
AQI (0-500): 432

Day 5
Temperature (10-50 C): 34
Humidity (0-100%): 23
AQI (0-500): 456

Day 6
Temperature (10-50 C): 12
Humidity (0-100%): 34
AQI (0-500): 123

Day 7
Temperature (10-50 C): 10
Humidity (0-100%): 54
AQI (0-500): 109

--- Enter readings for Hostel ---

Day 1
Temperature (10-50 C): 10
```

26°C
Partly cloudy

```
Humidity (0-100%): 78
AQI (0-500): 399

Day 3
Temperature (10-50 C): 12
Humidity (0-100%): 54
AQI (0-500): 123

Day 4
Temperature (10-50 C): 43
Humidity (0-100%): 29
AQI (0-500): 194

Day 5
Temperature (10-50 C): 23
Humidity (0-100%): 98
AQI (0-500): 187

Day 6
Temperature (10-50 C): 11
Humidity (0-100%): 123
Invalid humidity. Try again.
Humidity (0-100%): 89
AQI (0-500): 143

Day 7
Temperature (10-50 C): 34
Humidity (0-100%): 78
AQI (0-500): 432

--- Enter readings for Library ---

Day 1
Temperature (10-50 C): 11
Humidity (0-100%): 21
AQI (0-500): 234

Day 2
Temperature (10-50 C): 12
Humidity (0-100%): 31
AQI (0-500): 345

Day 3
Temperature (10-50 C): 13
Humidity (0-100%): 41
AQI (0-500): 454

Day 4
Temperature (10-50 C): 14
Humidity (0-100%): 51
AQI (0-500): 112

Day 5
Temperature (10-50 C): 15
Humidity (0-100%): 61
AQI (0-500): 183

Day 6
```

26°C

```
Temperature (10-50 C): 16
Humidity (0-100%): 71
AQI (0-500): 176

Day 7
Temperature (10-50 C): 17
Humidity (0-100%): 81
AQI (0-500): 164

--- Enter readings for Cafeteria ---

Day 1
Temperature (10-50 C): 11
Humidity (0-100%): 78
AQI (0-500): 431

Day 2
Temperature (10-50 C): 12
Humidity (0-100%): 68
AQI (0-500): 432

Day 3
Temperature (10-50 C): 13
Humidity (0-100%): 58
AQI (0-500): 433

Day 4
Temperature (10-50 C): 14
Humidity (0-100%): 48
AQI (0-500): 434

Day 5
Temperature (10-50 C): 15
Humidity (0-100%): 38
AQI (0-500): 435

Day 6
Temperature (10-50 C): 16
Humidity (0-100%): 28
AQI (0-500): 436

Day 7
Temperature (10-50 C): 17
Humidity (0-100%): 18
AQI (0-500): 437

--- Enter readings for Sports Area ---

Day 1
Temperature (10-50 C): 14
Humidity (0-100%): 76
AQI (0-500): 24

Day 2
Temperature (10-50 C): 66
Invalid temperature. Try again.
Temperature (10-50 C): 34
Humidity (0-100%): 453
```

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```
Day 3
Temperature (10-50 C): 15
Humidity (0-100%): 78
AQI (0-500): 324

Day 4
Temperature (10-50 C): 43
Humidity (0-100%): 76
AQI (0-500): 321

Day 5
Temperature (10-50 C): 11
Humidity (0-100%): 28
AQI (0-500): 111

Day 6
Temperature (10-50 C): 32
Humidity (0-100%): 26
AQI (0-500): 112

Day 7
Temperature (10-50 C): 45
Humidity (0-100%): 34
AQI (0-500): 115

All readings saved successfully.

========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
Enter your choice: 2

TEMPERATURE MATRIX (C)
Zone                 Day 1     Day 2     Day 3     Day 4     Day 5     Day 6     Day 7
Academic Block        12.0      23.0      16.0      45.0      34.0      12.0      [illegible]
Hostel                10.0      35.0      12.0      43.0      23.0      11.0      [illegible]
Library               11.0      12.0      13.0      14.0      15.0      16.0      17.0
Cafeteria             11.0      12.0      13.0      14.0      15.0      16.0      17.0
Sports Area           14.0      34.0      15.0      43.0      11.0      32.0      45.0

HUMIDITY MATRIX (%)
Zone                 Day 1     Day 2     Day 3     Day 4     Day 5     Day 6     Day 7
Academic Block        23.0      34.0      34.0      32.0      23.0      34.0      54.0
Hostel                54.0      78.0      54.0      29.0      98.0      89.0      78.0
Library               21.0      31.0      41.0      51.0      61.0      71.0      81.0
Cafeteria             78.0      68.0      58.0      48.0      38.0      28.0      18.0
Sports Area           76.0      43.0      78.0      76.0      26.0      26.0      34.0

AQI MATRIX
Zone                 Day 1     Day 2     Day 3     Day 4     Day 5     Day 6     Day 7
Academic Block          94       456       432       432       456       123       189
Hostel                 437       399       123       194       187       143       432
```

26°C
Partly cloudy

Search

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```
========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
Enter your choice: 3

ZONE-WISE AVERAGES
Zone                 Temp (C)     Humidity          AQI
Academic Block          21.71        33.43       291.71
Hostel                  24.00        68.57       273.57
Library                 14.00        51.00       238.29
Cafeteria               14.00        48.00       434.00
Sports Area             27.71        51.57       208.57

========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
Enter your choice: 4

RISK DAY REPORT

Academic Block:
Risk days: 4
Risk detected on: Day 2 Day 3 Day 4 Day 5

Hostel:
Risk days: 6
Risk detected on: Day 1 Day 2 Day 4 Day 5 Day 6 Day 7

Library:
Risk days: 6
Risk detected on: Day 1 Day 2 Day 3 Day 5 Day 6 Day 7

Cafeteria:
Risk days: 7
Risk detected on: Day 1 Day 2 Day 3 Day 4 Day 5 Day 6 Day 7

Sports Area:
Risk days: 4
Risk detected on: Day 2 Day 3 Day 4 Day 7

========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
```

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```
Enter your choice: 5

ZONE-WISE AVERAGES
Zone                 Temp (C)     Humidity          AQI
Academic Block          21.71        33.43       291.71
Hostel                  24.00        68.57       273.57
Library                 14.00        51.00       238.29
Cafeteria               14.00        48.00       434.00
Sports Area             27.71        51.57       208.57

GREEN INDEX REPORT
Academic Block      Green Index: 90.55
Hostel              Green Index: 68.17
Library             Green Index: 93.54
Cafeteria           Green Index: 79.68
Sports Area         Green Index: 79.11

ZONE RANKING (BEST TO WORST)
Rank     Zone                 Green Index
1        Library              93.54
2        Academic Block       90.55
3        Cafeteria            79.68
4        Sports Area          79.11
5        Hostel               68.17

========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
Enter your choice: 6

ZONE-WISE AVERAGES
Zone                 Temp (C)     Humidity          AQI
Academic Block          21.71        33.43       291.71
Hostel                  24.00        68.57       273.57
Library                 14.00        51.00       238.29
Cafeteria               14.00        48.00       434.00
Sports Area             27.71        51.57       208.57

RISK DAY REPORT

Academic Block:
Risk days: 4
Risk detected on: Day 2 Day 3 Day 4 Day 5

Hostel:
Risk days: 6
Risk detected on: Day 1 Day 2 Day 4 Day 5 Day 6 Day 7

Library:
Risk days: 6
Risk detected on: Day 1 Day 2 Day 3 Day 5 Day 6 Day 7

Cafeteria:
```

26°C
Search

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```
Risk detected on: Day 1 Day 2 Day 4 Day 5 Day 6 Day 7

Library:
Risk days: 6
Risk detected on: Day 1 Day 2 Day 3 Day 5 Day 6 Day 7

Cafeteria:
Risk days: 7
Risk detected on: Day 1 Day 2 Day 3 Day 4 Day 5 Day 6 Day 7

Sports Area:
Risk days: 4
Risk detected on: Day 2 Day 3 Day 4 Day 7

ENVIRONMENTAL RECOMMENDATIONS

Academic Block:
- Reduce vehicle idling and improve ventilation.
- Monitor environmental conditions regularly.

Hostel:
- Reduce vehicle idling and improve ventilation.
- Monitor environmental conditions regularly.

Library:
- Reduce vehicle idling and improve ventilation.
- Monitor environmental conditions regularly.

Cafeteria:
- Reduce vehicle idling and improve ventilation.
- Monitor environmental conditions regularly.

Sports Area:
- Reduce vehicle idling and improve ventilation.
- Monitor environmental conditions regularly.

========== ECOGUARD MENU ==========
1. Input / Update Data
2. Display Data
3. Zone-wise Averages
4. Detect Risk Days
5. Green Index and Ranking
6. Recommendations
7. Exit
Enter your choice: 7

Exiting EcoGuard. Goodbye!
yokshita@yokshithachowdary:~$
```